

Class Imbalance and Its Impact on Interpreting Errors

Intro

By the end, you'll know:

- What class imbalance is
- Why accuracy can be misleading
- How to evaluate models properly

Concept Explanation

Class imbalance simply means — one class has **many more samples** than another.

You'll see this in real-life ML datasets all the time:

- Spam detection — most emails are *not spam*
- Disease prediction — most people are *not sick*
- Credit card fraud — most transactions are *safe*

So when the dataset is **90% one type and 10% another**, the model starts to get lazy.

It just learns to always predict the majority class and still looks good because it's "right" most of the time.

Accuracy loves imbalance because true negatives dominate the score.

That's why we need other metrics like:

- **Precision:** Out of the predicted positives, how many are correct?
- **Recall (Sensitivity):** Out of the actual positives, how many did we catch?

Visual & Mathematical Explanation

Here's how imbalance fools the confusion matrix.

Think of this 2x2 grid:

	Predicted Positive	Predicted Negative
Actual Positive	TP (Correct detection)	FN (Missed case)
Actual Negative	FP (False alarm)	TN (Correct rejection)

When your dataset has mostly negatives — say, 990 "No Fraud" and only 10 "Fraud" — most cells in the matrix will fill up with **TN (True Negatives)**.

That means accuracy will look high even if the model misses *every* fraud case.
So, we shift focus from *overall accuracy* to *error types*.

In imbalanced problems, **False Negatives hurt more**.

Imagine a cancer test marking sick patients as healthy — that's a critical miss.

Mini Real-World Use Cases

Example 1 – Medical Diagnosis:

You're building a model to detect a rare disease.

Only 2% of patients are sick.

If your model says "healthy" for everyone, you get 98% accuracy — but you've failed all the real patients.

In healthcare, **recall** is more important — catching as many actual positives as possible.

Example 2 – Spam Detection:

In spam filters, it's okay if you sometimes mark a real email as spam (false positive).

It's annoying, but not dangerous.

So here, **precision** might matter more — we don't want to block too many legit emails.

The lesson?

👉 Different problems require different interpretations of model errors.

Summary/ Reflection

Alright, let's wrap this up.

What we learned today:

- **Class imbalance** means one class dominates the dataset.
- **Accuracy** can lie — always check **precision**, **recall**, and **F1-score**.
- **Context matters** — in healthcare, recall may matter more; in spam detection, precision.
- You can handle imbalance with techniques like **resampling**, **class weights**, or **threshold tuning** — which we'll explore later.